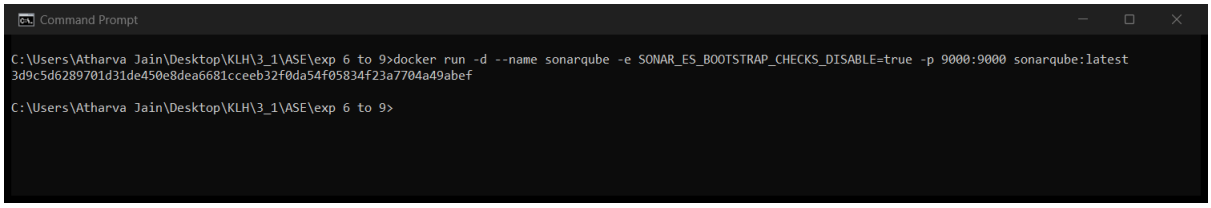
ASE LAB 8

Step 1:

- Firstly Open the docker desktop
- Then run this command:

    docker run -d --name sonarqube -e SONAR_ES_BOOTSTRAP_CHECKS_DISABLE=true -p 9000:9000 sonarqube:latest



- Wait until the command and then run: docker ps

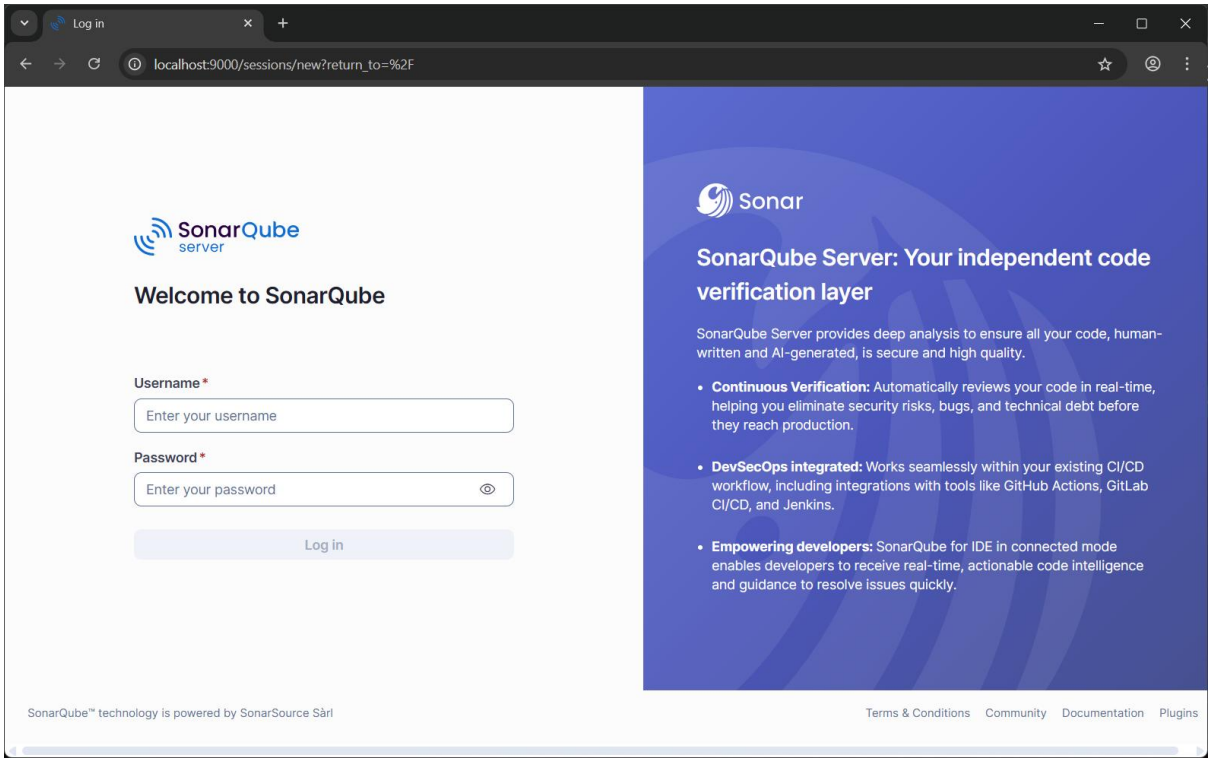
    If u see output like this:



Then The image has been installed successfully

Step 2: Open sonarqube using localhost

http://localhost:9000

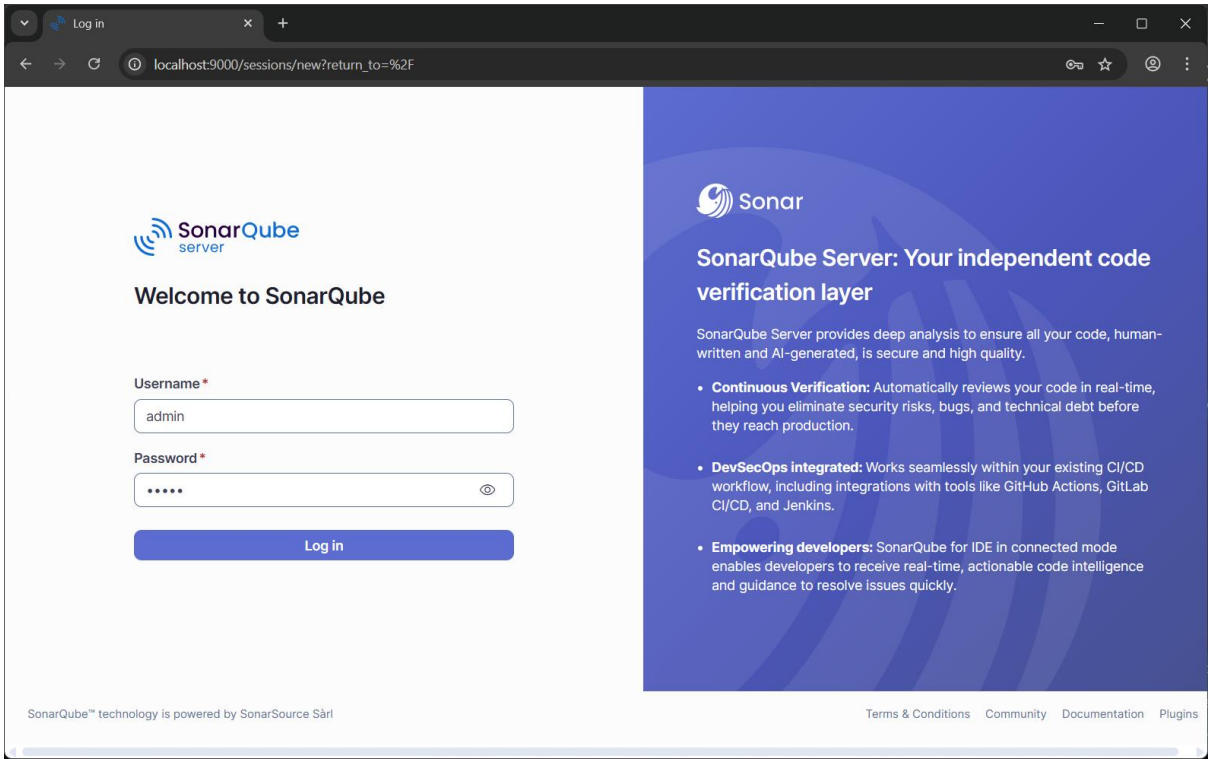


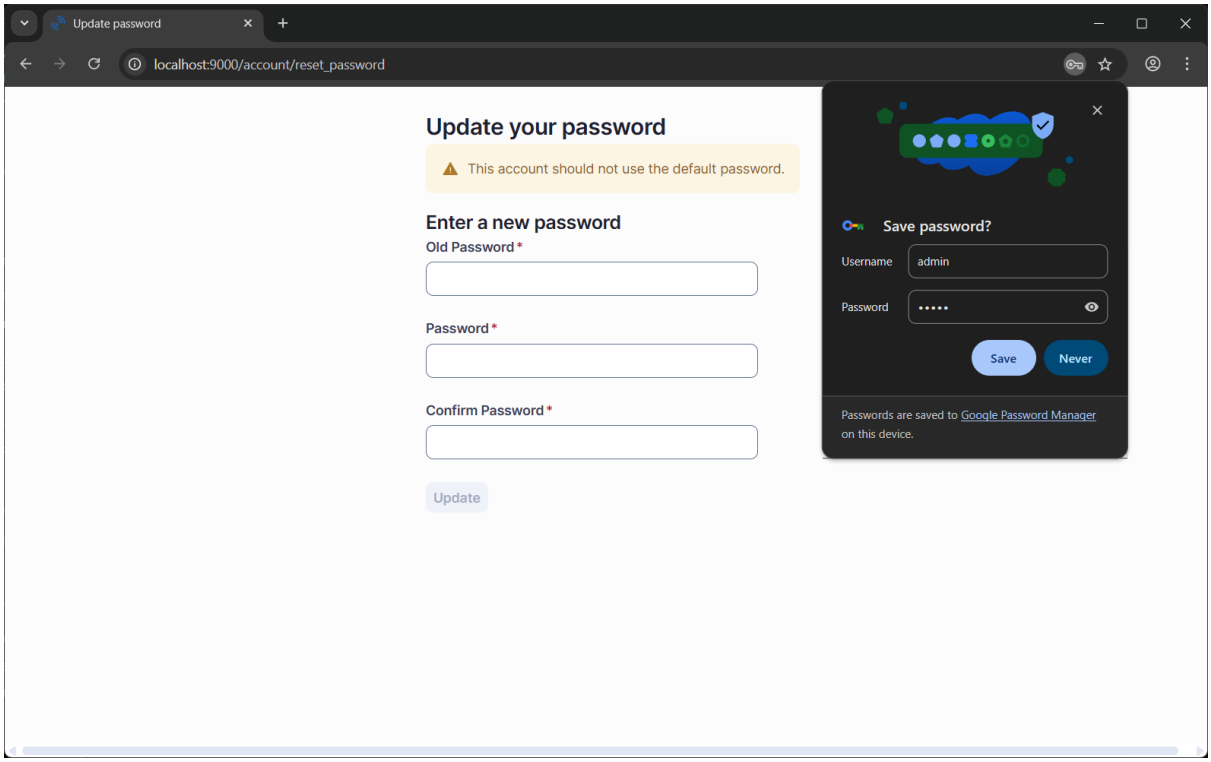
➔ Then login using the credentials:

Username: admin

Password: admin

On the first login SonarQube asks you to update the default admin password. Enter the old password (admin) and a new password, then click Update.

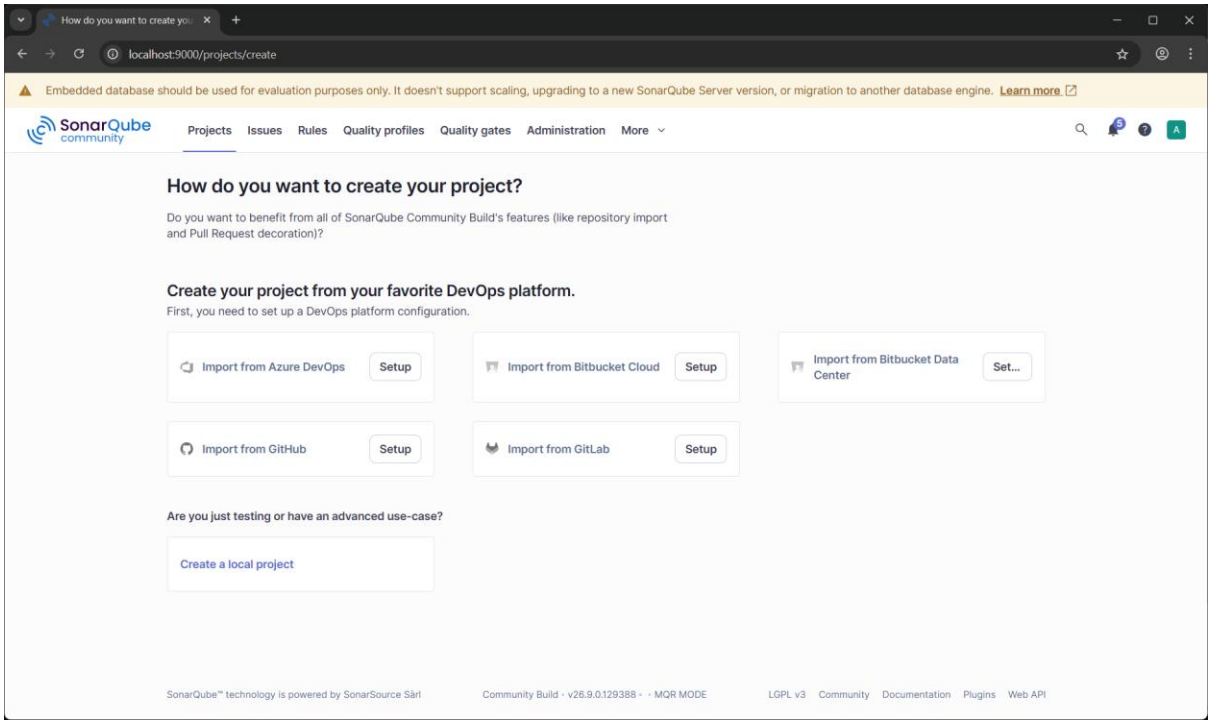


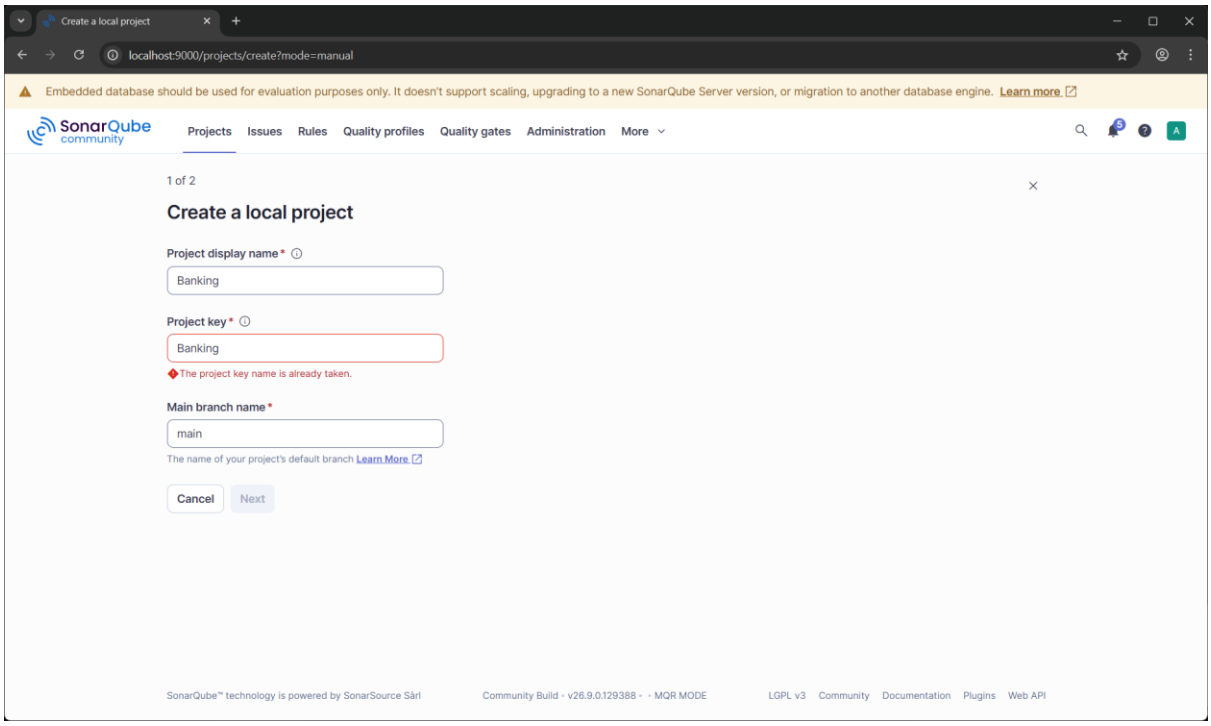


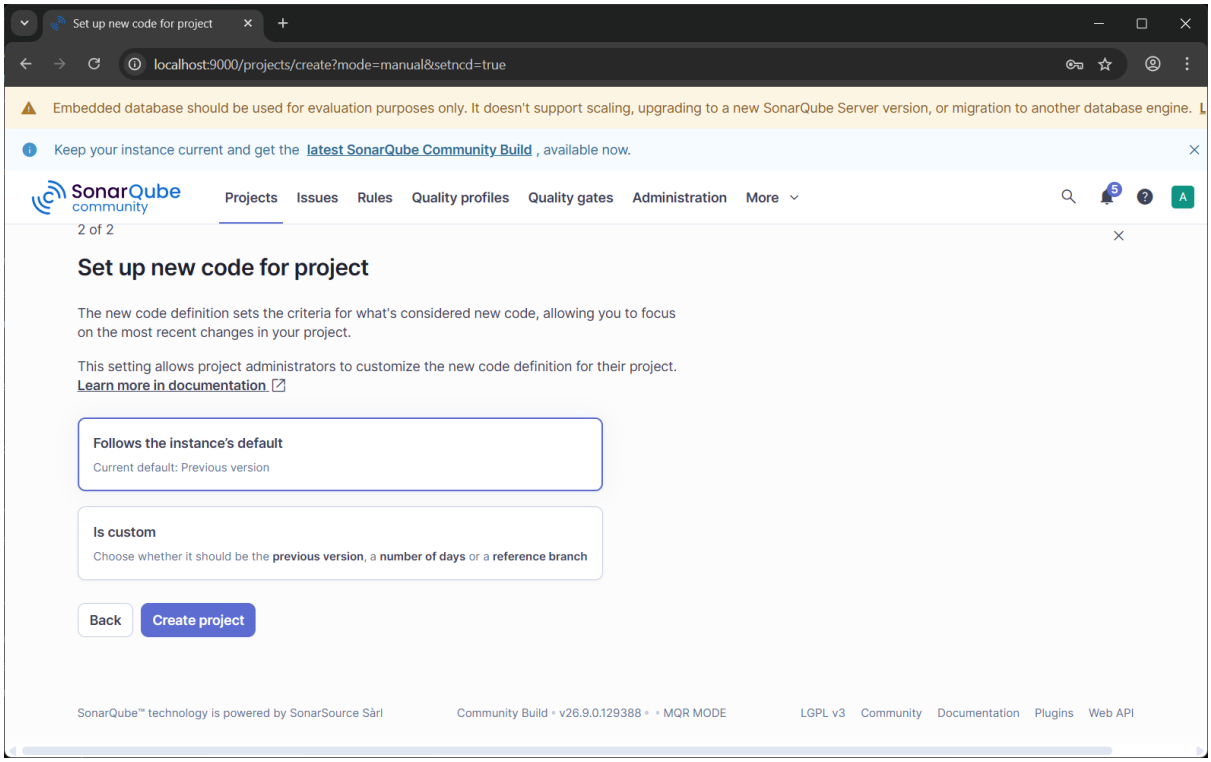
Step 3: Create the SonarQube project

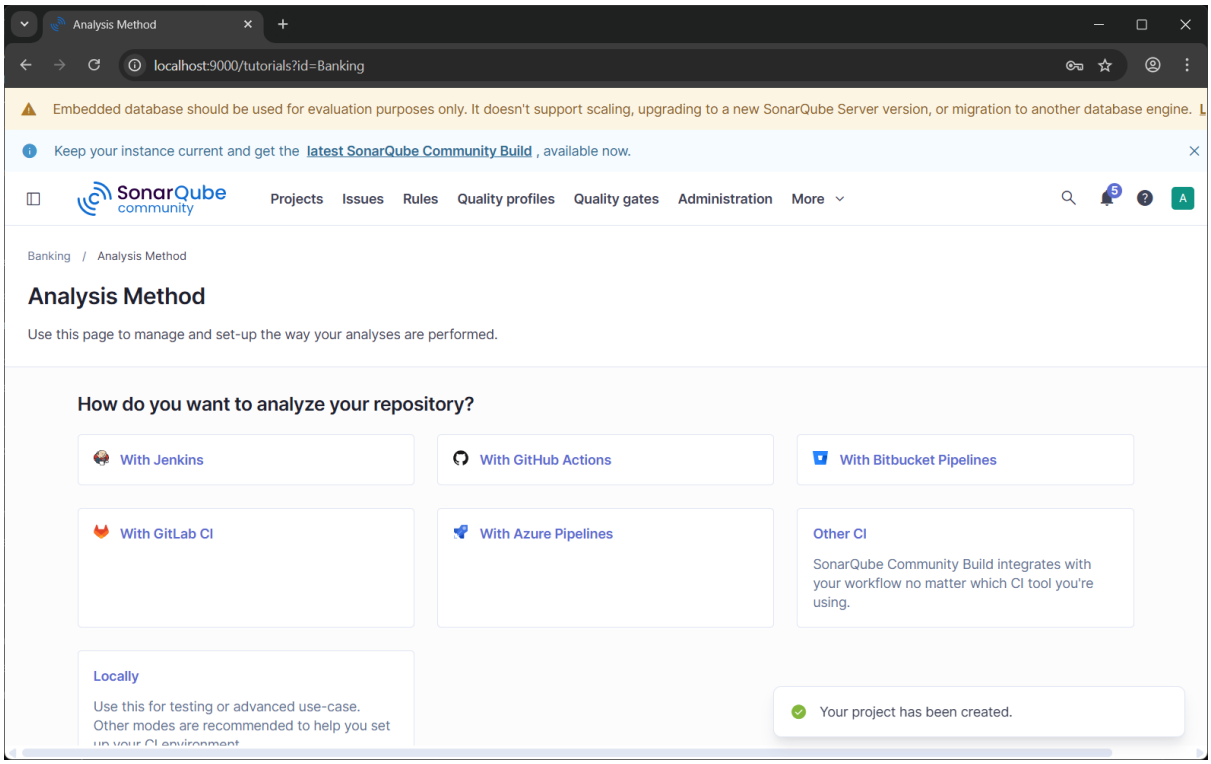
Project Name: Banking

Project Key: Banking


How do you want to create you
localhost:9000/projects/create
Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. Learn more
SonarQube community
Projects Issues Rules Quality profiles Quality gates Administration More
How do you want to create your project?
Do you want to benefit from all of SonarQube Community Build's features (like repository import and Pull Request decoration)?
Create your project from your favorite DevOps platform.
First, you need to set up a DevOps platform configuration.
Import from Azure DevOps Setup
Import from Bitbucket Cloud Setup
Import from Bitbucket Data Center Set...
Import from GitHub Setup
Import from GitLab Setup
Are you just testing or have an advanced use-case?
Create a local project
SonarQube™ technology is powered by SonarSource Sàrl
Community Build · v26.9.0.129388 · · MQR MODE
LGPL v3 Community Documentation Plugins Web API


Create a local project
localhost:9000/projects/create?mode=manual
Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. Learn more
SonarQube community
Projects Issues Rules Quality profiles Quality gates Administration More
1 of 2
Create a local project
Project display name *
Banking
Project key *
Banking
The project key name is already taken.
Main branch name *
main
The name of your project's default branch Learn More
Cancel Next
SonarQube™ technology is powered by SonarSource Sàrl
Community Build · v26.9.0.129388 · · MQR MODE
LGPL v3 Community Documentation Plugins Web API





STEP 4 — Create your Java project folder

The structure should be like this:

ASE 8

│

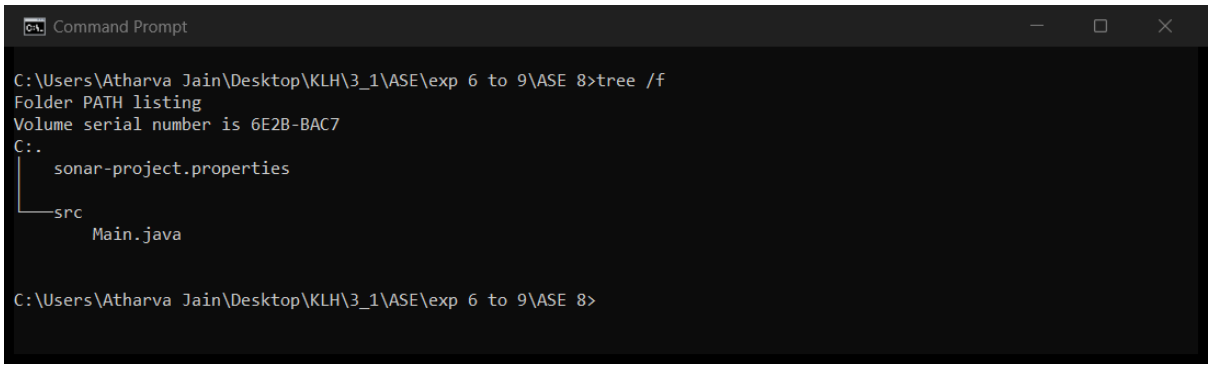
├── sonar-project.properties

│

└── src

    │

    └── Main.java



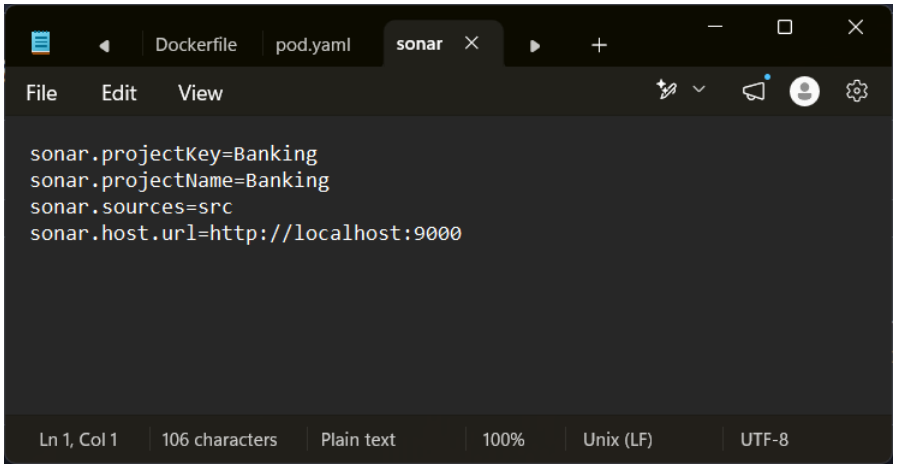
sonar-project.properties:

sonar.projectKey=Banking

sonar.projectName=Banking

sonar.sources=src

sonar.host.url=http://localhost:9000



Main.java: (can write your own code if not just copy paste this):

```
public class Main {

    public static void main(String[] args) {

        int a = 10;
        int b = 20;

        System.out.println("Result: " + calculate(a, b));

        String name = null;

        if (name != null) {
            System.out.println(name);
        }

        int unused = 100;

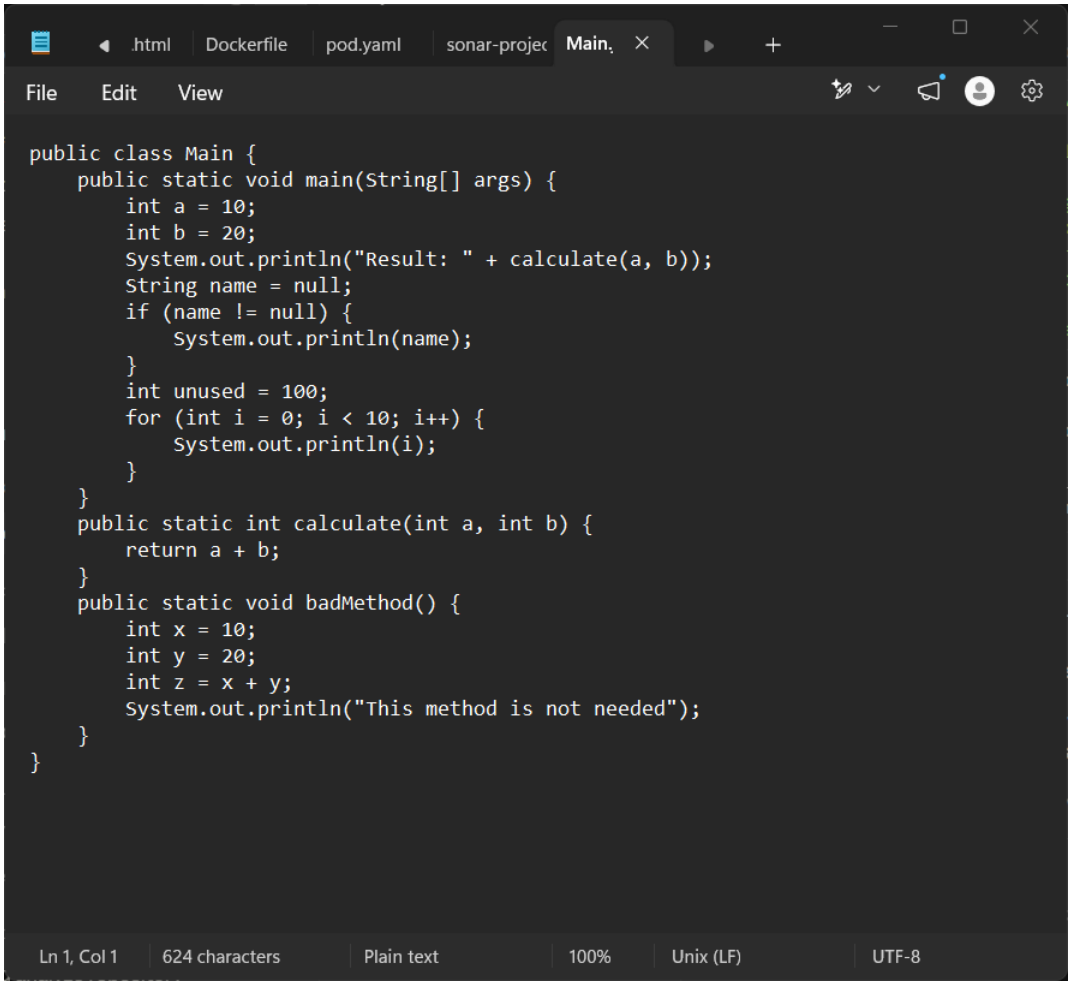
        for (int i = 0; i < 10; i++) {
            System.out.println(i);
        }
    }

    public static int calculate(int a, int b) {
        return a + b;
    }

    public static void badMethod() {
        int x = 10;
        int y = 20;
```

```
        int z = x + y;

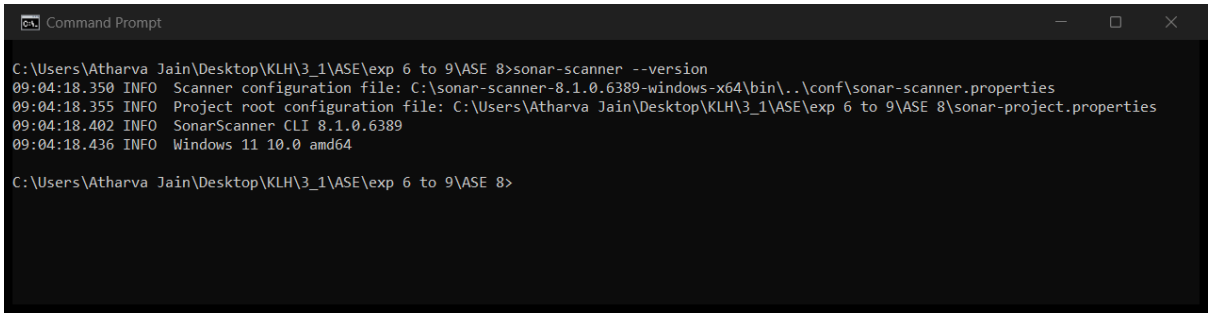
        System.out.println("This method is not needed");
    }
}
```



Step 5: Install SonarScanner

https://docs.sonarsource.com/sonarqube-server/analyzing-source-code/scanners/sonarscanner

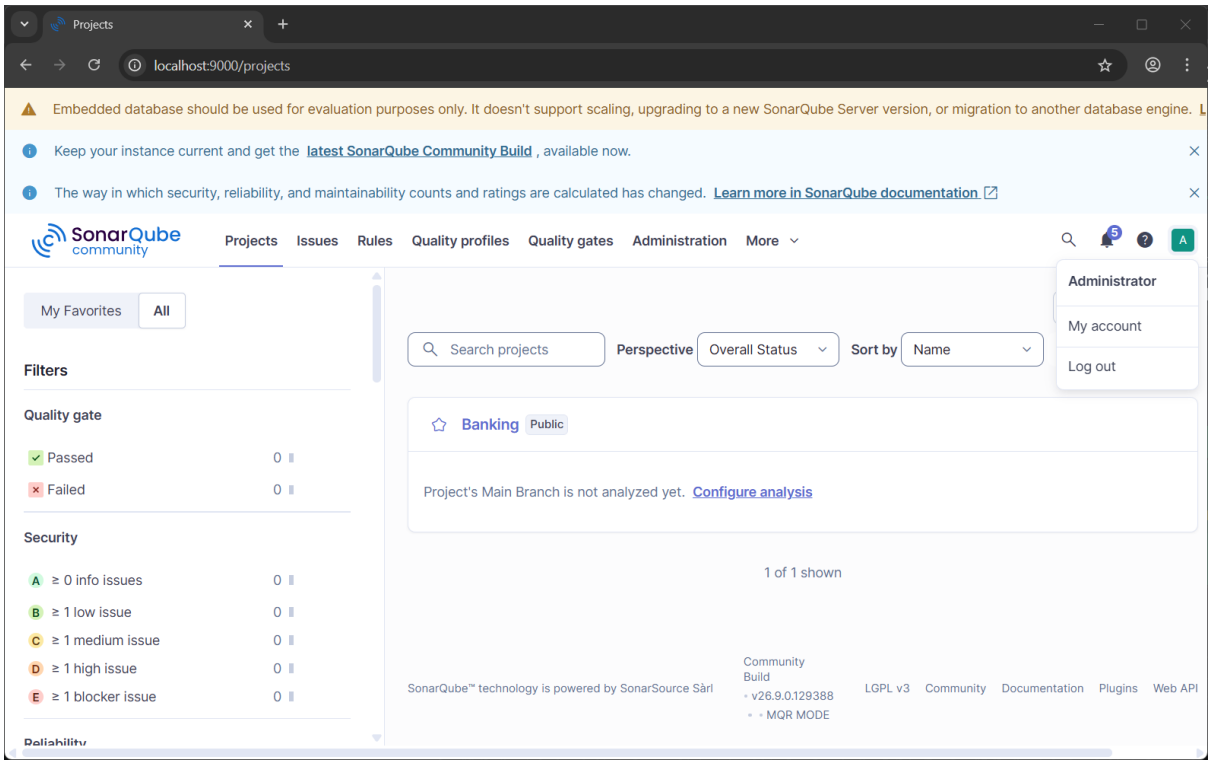
➔ To check if its installed correctly use: sonar-scanner –version



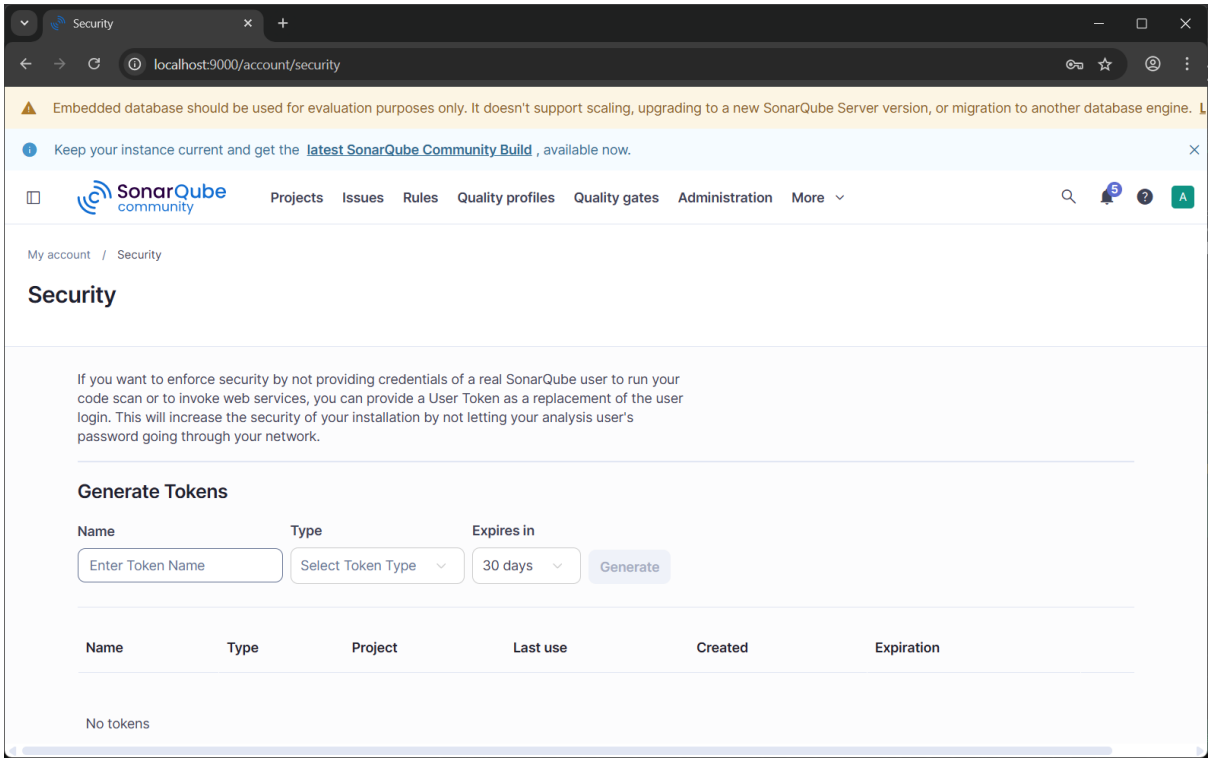
Step 6: Generate SonarQube authentication token

➔ Go to the localhost:9000, login using the credentials

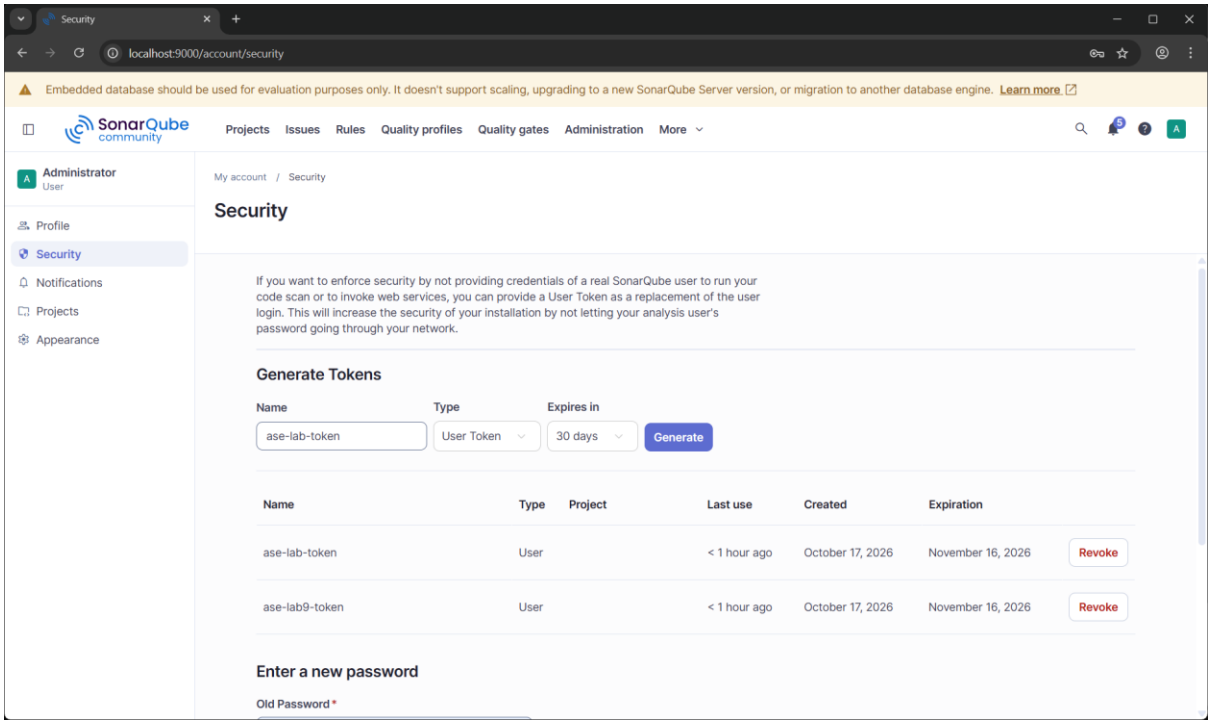
➔ After logging in open your profile on the right corner

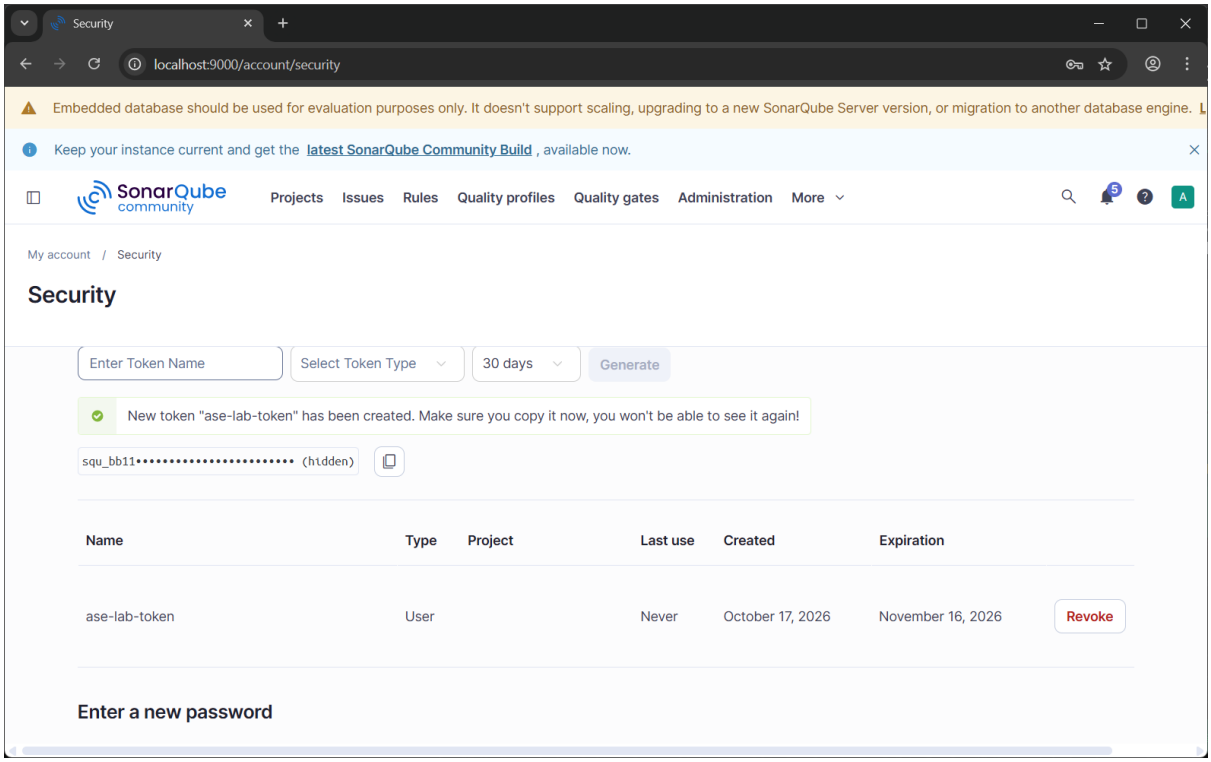


➔ Then go to the Security option in the left



➔ Give a name to the token and then select the type as User token and generate it





➔ Copy the token and open the cmd and run

set SONAR_TOKEN=YOUR_TOKEN

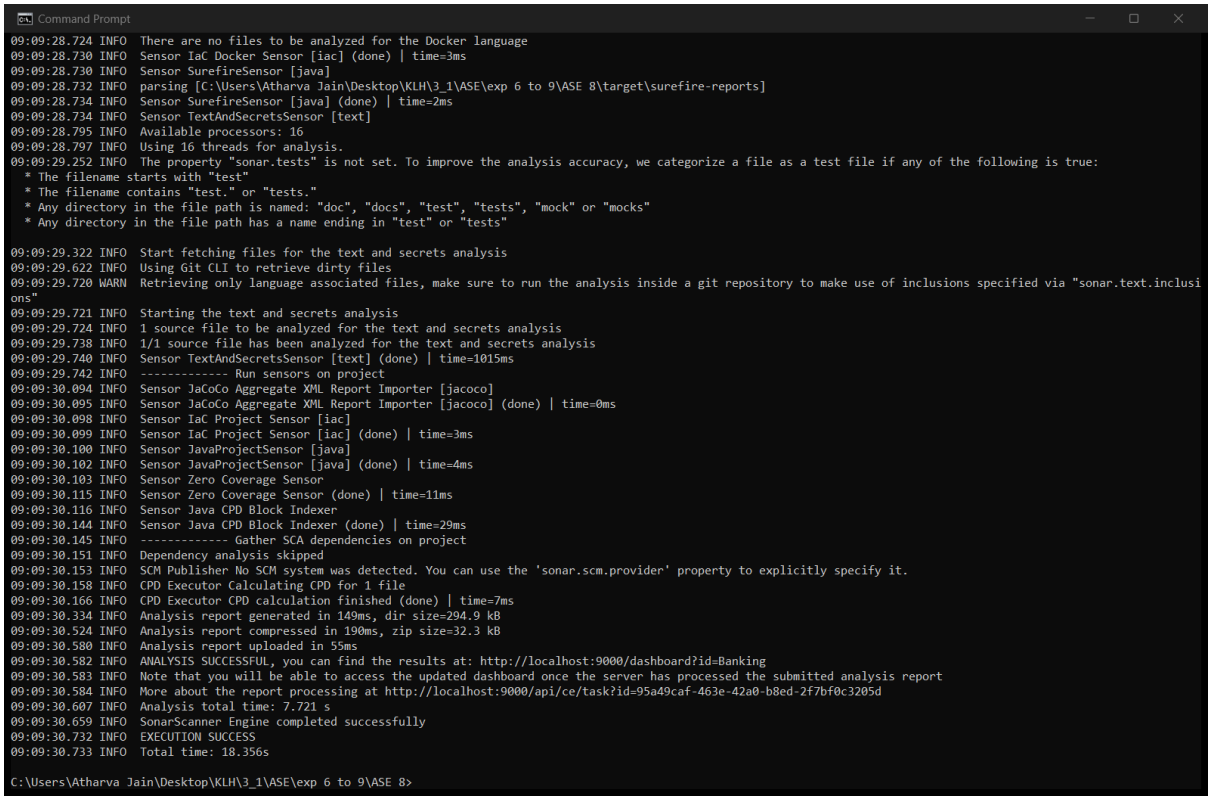
Step 7: Go to your project directory

➔ Open cmd and then go to your project directory ASE 8

Step 8: Run SonarScanner

➔ After going to your directory in cmd run the command:

sonar-scanner



➔ When you see the execution as successful go the localhost:9000

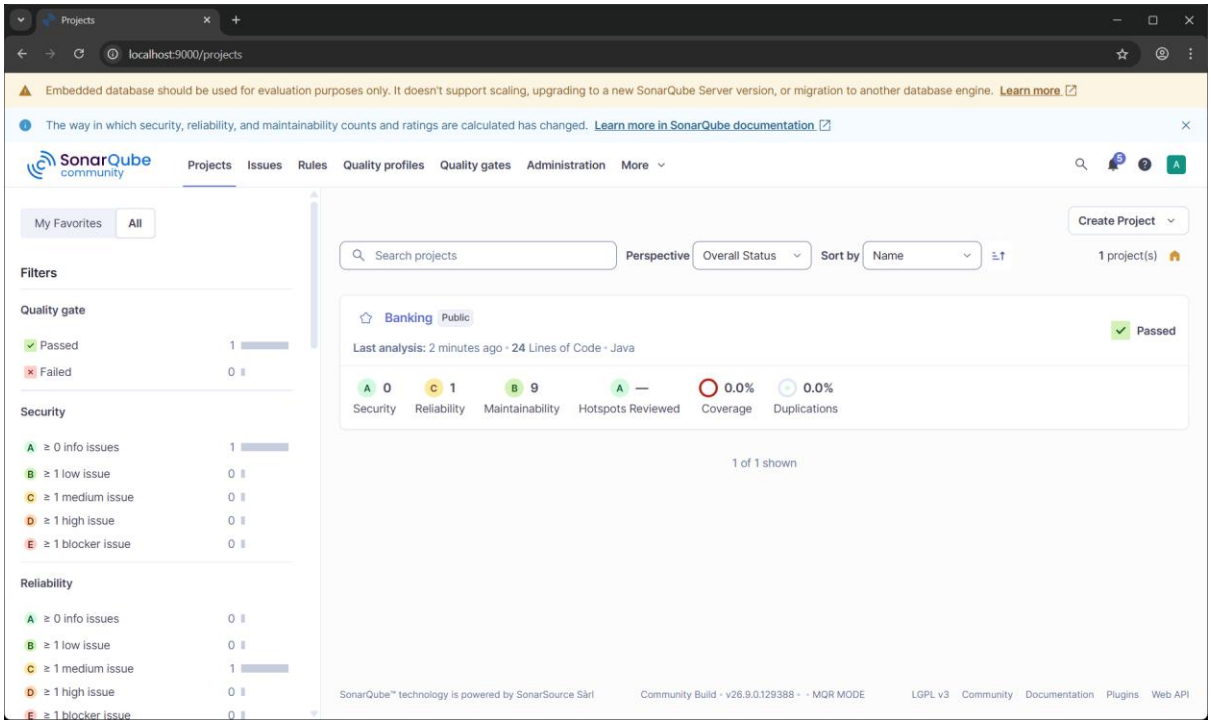
Step 9: Open your project dashboard

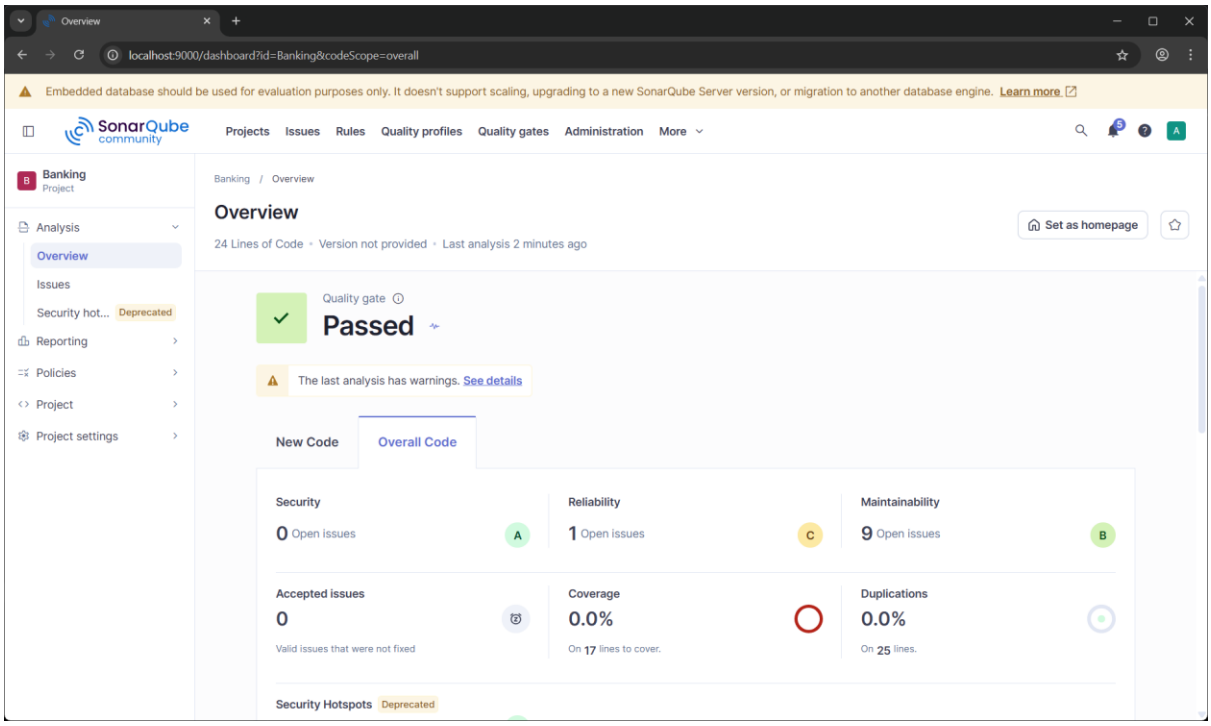
Go to:

http://localhost:9000

Then:

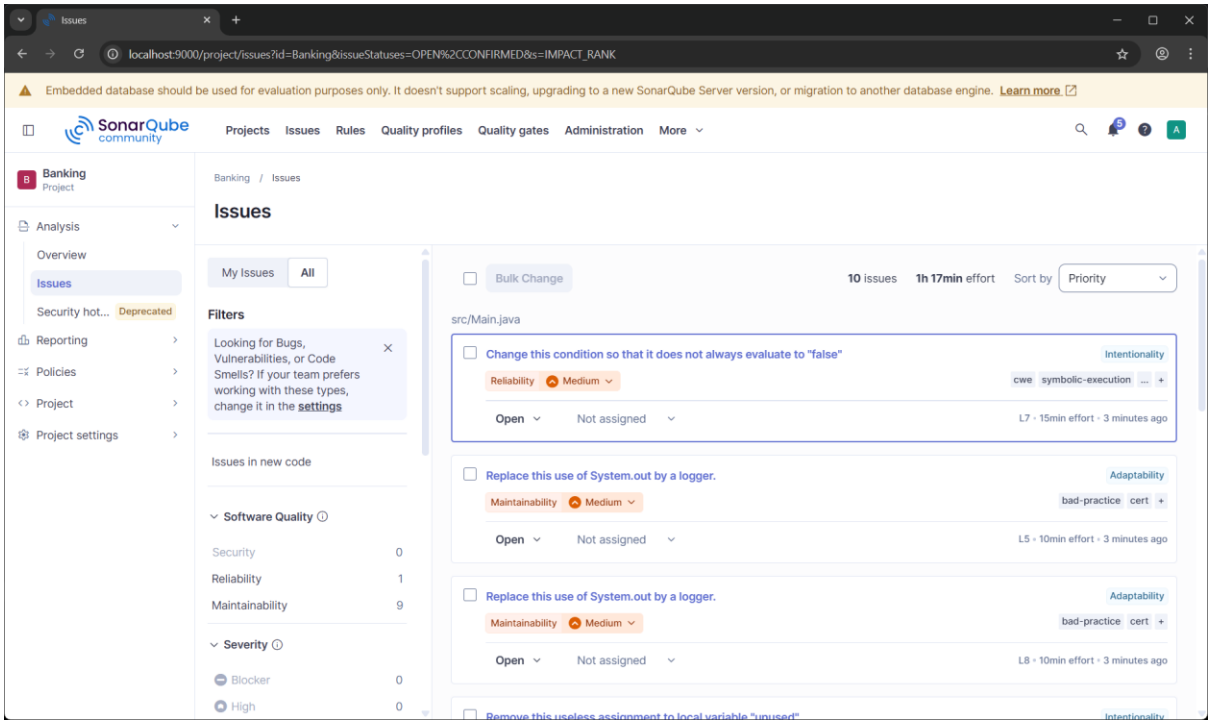
Projects → Banking





Here u should see your code if its passed or failed

Result on this system: Quality Gate Passed — 24 lines of code, 0 security issues, 1 reliability issue, 9 maintainability issues, 0.0% duplications (10 issues in total).


Issues
localhost:9000/project/issues?id=Banking&issueStatuses=OPEN%2CCONFIRMED&s=IMPACT_RANK
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Project
Analysis
Overview
Issues
Security hot... Deprecated
Reporting
Policies
Project
Project settings
Banking / Issues
Issues
My Issues All
Filters
Looking for Bugs, Vulnerabilities, or Code Smells? If your team prefers working with these types, change it in the settings
Issues in new code
Software Quality
Security 0
Reliability 1
Maintainability 9
Severity
Blocker 0
High 0
Bulk Change
10 issues 1h 17min effort Sort by Priority
src/Main.java
Change this condition so that it does not always evaluate to "false"
Intentionality
Reliability Medium
cwe symbolic-execution ... +
Open Not assigned
L7 • 15min effort • 3 minutes ago
Replace this use of System.out by a logger.
Adaptability
Maintainability Medium
bad-practice cert +
Open Not assigned
L5 • 10min effort • 3 minutes ago
Replace this use of System.out by a logger.
Adaptability
Maintainability Medium
bad-practice cert +
Open Not assigned
L8 • 10min effort • 3 minutes ago
Remove this useless assignment to local variable "unused"
Intentionality